# Molded Pulp for Oji India: Segment Prioritisation and Pilot Proposal

*Independent desk-research project by Jayanth Ram, IIT Hyderabad. 25 October 2026. Companion files: OJI_Project.xlsx (all numbers and formulas) and the five-slide summary.*

## 1. Objective

Oji Group's India business-development arm is looking at Indian demand for paper, packaging and forest-based products. Oji's own packaging page lists Molded Pulp as a product, described as having a secure fit, leak-proof performance, an exceptionally smooth finish, FSC certification, and biodegradable and compostable properties (S01). That page does not mention India.

**Research question:** Which one of three Indian packaging segments (electronics and e-commerce protective packaging, food and FMCG packaging, or automotive and industrial components) should Oji India prioritise for a molded-pulp pilot, and what must a six-month pilot prove?

**Short answer:** Food and FMCG packaging, narrowly. It leads on base weights (3.30 vs 3.20 for electronics) because it combines the largest reported demand with the best fit to Oji's stated product claims. The lead is small and reverses if feasibility is weighted heavily, so the pilot is designed to test the assumption that decides the outcome: whether Indian buyers will pay a premium for fibre packaging with barrier and leak-proof performance. Automotive and industrial components trail clearly.

## 2. Method and evidence quality

This is desk research. I logged 13 sources in a Source Log, each scored 1 to 5 for reliability, with the specific claim used and its limitations. I benchmarked eight competitors or alternatives, scored the three segments on six weighted criteria, and built low, base and high market-sizing scenarios.

Three points about evidence quality shape everything below.

- **No primary data.** I did not interview buyers, obtain prices, or test samples. Every conclusion is provisional.
- **The two market-size sources conflict.** A research vendor puts India's molded pulp market at USD 291 million in 2025, with food about 52% of it (S07), which is roughly USD 151 million. The same vendor family puts India's molded-fibre protective packaging for electronics alone at USD 0.17 billion (S08). Food plus electronics then exceeds the total. Both cannot be right as stated, so I treat them as a range and rate both 2 out of 5.
- **Some inputs are assumptions.** Where no source exists (the auto share, the electronics low case, the addressable and obtainable shares), I used labelled assumptions in yellow cells. They are flagged as such wherever they appear.

Confidence tags used in the text: **[Certain]** hard evidence, **[Likely]** strong inference, **[Guessing]** gap-filling.

## 3. Findings

### 3.1 What Oji is offering

Oji's page (S01) positions molded pulp alongside two other fibre-based products: a Walki fibre lid that replaces aluminium lids with gas, water-vapour and aroma barrier properties, and a 100% fibre stand-up pouch that is plastic-free and heat-sealable. The page frames the offer around the EU Packaging and Packaging Waste Regulation, noting that all packaging on the EU market must be designed to be recyclable by January 2030. **[Certain]** these are Oji's own claims. **[Certain]** the page gives no price, load rating, barrier data, food-contact certification or India availability.

Oji's internship brief also lists "Sustainable Package (Biodegradable, High Barrier Function)" as an ongoing India project. **[Likely]** this makes consumer and food-facing packaging the closest match between what Oji already sells and what its India team is exploring.

### 3.2 The Indian molded pulp market

The vendor figures are the only market-size numbers I found, and they are weak (S07, S08).

| Item | Figure | Source | Reliability |
|---|---|---|---|
| India molded pulp market, 2025 | USD 291M | S07 | 2 |
| Forecast, 2035 | USD 721.16M (9.5% CAGR) | S07 | 2 |
| Food packaging share, 2025 | about 52% | S07 | 2 |
| Trays share, 2025 | about 41% | S07 | 2 |
| Electronics protective packaging, 2025 | USD 0.17bn (7.4% CAGR to 2035) | S08 | 2 |

**[Likely]** growth is real but the level is uncertain. The trade press also describes India's sector as less organised than China's and Western Europe's, which is consistent with patchy data.

### 3.3 Policy and demand drivers

- **Plastic regulation (S04).** India's Plastic Waste Management Rules, through the CPCB's EPR guidelines notified on 16 February 2022, oblige producers, importers and brand owners to register on a central portal and meet EPR obligations for plastic packaging. **[Certain]** the rules cover plastic. **[Likely]** they raise the cost of plastic packaging and so create indirect pull for alternatives. **[Certain]** they do not mandate fibre.
- **Alternatives to plastics (S05).** NITI Aayog's 2022 report on alternatives to plastics covers substitutes, technology readiness and a roadmap. It is four years old, and I have not confirmed that it discusses molded pulp.
- **Electronics manufacturing (S02).** The Union Budget 2026-27 raised the Electronics Components Manufacturing Scheme outlay to INR 40,000 crore. The government expects INR 29,024 crore of scheme production by the end of FY2026-27. **[Certain]** this is a scheme projection, not packaging demand.

- **Automotive components (S03).** ACMA reports FY2025-26 turnover of INR 7.60 lakh crore (USD 85.9 billion), up 12.7%. ACMA's own About page shows INR 7.59 lakh crore, a rounding mismatch I note rather than resolve. **[Certain]** this shows sector scale. **[Guessing]** packaging intensity and fibre suitability for auto parts are unproven.

### 3.4 Competitors and alternatives

I benchmarked eight. Evidence confidence is low to moderate throughout, because most entries rest on company self-descriptions or vendor summaries.

| Competitor | What sources show | Segment | Confidence |
|---|---|---|---|
| Maspack Limited (Hyderabad) | Industrial pulp trays and fruit trays; machines for egg and apple trays; ISO 9001 and CE listed (S09) | Electronics, industrial | 3 |
| Pulp2Pack (New Delhi) | Trays and corner guards; electronics use cited (S10) | Electronics | 3 |
| S M Pulp Packaging | Auto, medical, electronics, household range (S12) | Auto, electronics | 3 |
| Ecology Pac LLP (UP) | Pharma, cosmetic and egg trays (S06) | Niche trays | 3 |
| Huhtamaki PPL | Fibre trays and components for food, beverage, personal care (S07) | Food / FMCG | 2 |
| Pakka and Neeyog | Compostable food serviceware, meal trays, bagasse containers (S07) | Food service | 2 |
| Ecosure Pulpmolding | Made-in-India automatic pulp molding machines (S11) | Enabler | 3 |
| Plastic, EPS, corrugated | No cost source gathered; plastic faces EPR (S04) | All | 1 |

Three observations follow. First, **[Likely]** local makers already supply electronics trays, so electronics is not a green field. Second, **[Likely]** food is the more crowded space, with large and specialised incumbents. Third, **[Likely]** domestic machine makers such as Ecosure lower the cost of entry for local competitors, which weakens any technology-only advantage. The same fact points to a possible local-production route for Oji.

## 4. Segment comparison

I scored each segment from 1 (weak) to 5 (strong) on six criteria. Scores are my judgement, tied to the evidence in the workbook's rationale columns.

| Criterion | Base weight | Electronics | Food / FMCG | Auto / industrial |
|---|---|---|---|---|

| Criterion | Base weight | Electronics | Food / FMCG | Auto / industrial |
|---|---|---|---|---|
| Demand potential | 20% | 4 | 4 | 3 |
| Customer fit with Oji's offer | 20% | 3 | 5 | 2 |
| Competitive gap | 15% | 3 | 2 | 3 |
| Adoption feasibility | 15% | 3 | 3 | 2 |
| Commercial feasibility | 15% | 3 | 2 | 2 |
| Evidence confidence | 15% | 3 | 3 | 2 |
| **Weighted score (base)** | | **3.20** | **3.30** | **2.35** |

**Sensitivity.** The ranking depends on weights.

| Weight set | Electronics | Food / FMCG | Auto / industrial | Leader |
|---|---|---|---|---|
| Base | 3.20 | 3.30 | 2.35 | Food / FMCG |
| Demand-heavy | 3.30 | 3.45 | 2.40 | Food / FMCG |
| Feasibility-heavy | 3.10 | 3.00 | 2.25 | Electronics |

A lead of 0.10 is within the noise of a single score moving by one point. The honest reading is: food and electronics are close, with food ahead on demand and fit, and electronics ahead on competitive gap and commercial feasibility. Automotive and industrial components are well behind in every weighting, mainly because I found no evidence that Oji's product fits the segment or that auto buyers are looking for fibre packaging **[Guessing]**.

## 5. Market sizing scenarios

The model runs from the 2025 segment size, grows it to 2028, applies an addressable share (premium, barrier-capable or spec-driven buyers) and an obtainable share (what a new entrant might win). The last two shares are unsourced assumptions, applied equally to every segment so that segment size, not a favourable assumption, drives the ranking.

| Segment, 2025 size (USD M) | Low | Base | High |
|---|---|---|---|
| Electronics / e-commerce | 29.1 | 99.6 | 170.0 |
| Food / FMCG | 122.2 | 151.3 | 180.4 |
| Auto / industrial | 5.8 | 14.6 | 23.3 |

How these were built: food is 42%, 52% or 62% of the S07 total, with 52% from S07 and the rest my +/- 10-point assumption. Electronics uses a 10% share of the S07 total for the low case, S08's USD 170M for the high case, and the midpoint for base. Auto uses a 2%, 5% or 8% placeholder share. In the high case

the three segments sum to 128% of S07's total, which confirms the sources conflict, so the scenarios are not additive.

| Obtainable revenue, 2028 | Low | Base | High |
|---|---|---|---|
| Electronics (USD M) | 0.04 | 0.39 | 1.86 |
| Food / FMCG (USD M) | 0.15 | 0.60 | 1.97 |
| Auto / industrial (USD M) | 0.01 | 0.06 | 0.25 |
| Electronics (INR crore) | 0.3 | 3.5 | 16.5 |
| Food / FMCG (INR crore) | 1.3 | 5.3 | 17.5 |
| Auto / industrial (INR crore) | 0.1 | 0.5 | 2.3 |

INR figures use 88.5 INR per USD, derived from ACMA's conversion (S03). The takeaway is not any single number. It is that the opportunity is modest (a few crore in the base case) and that the biggest swing factors are the unsourced funnel assumptions. That is exactly what a pilot should replace with buyer evidence.

## 6. Recommendation

**Prioritise food and FMCG packaging for the pilot, focused on premium and barrier-sensitive uses, and keep electronics and e-commerce protective packaging as the fast follower.**

Reasons:

1. **Best fit with what Oji already sells.** Leak-proof performance, compostability and FSC certification map to food buyers, and Oji's fibre lid and pouch (S01) extend the same story. The India project list points the same way.
2. **Largest reported demand.** Food is the biggest part of the reported market (S07), although that figure is low-reliability.
3. **Regulatory tailwind.** EPR obligations on plastic packaging (S04) make plastic costlier for brands **[Likely]**.

Trade-offs I am accepting:

- Food is crowded and price-sensitive. If Indian buyers will not pay a premium, this recommendation fails, and electronics becomes the better choice because its buyers are fewer and spec-driven.
- Food-contact compliance is a gate I have not researched **[Likely]** it will be required.
- The margin over electronics is 0.10 points. I would not defend this as a confident ranking. I would defend it as a reasoned starting point with a built-in test.

## 7. Proposed pilot (six months, January to June 2027)

**Candidate buyer types to validate (not named prospects):** premium ready-to-eat and meal brands; quick-service and food-delivery packaging buyers; packaged-food and personal-care brands facing plastic EPR costs.

| ID | Hypothesis | Test | Success metric | Kill criterion | Month |
|---|---|---|---|---|---|
| H1 | Buyers will pay a premium for barrier and leak-proof fibre packaging | 8 to 10 buyer interviews | At least 3 of 8 name an acceptable premium | Fewer than 2 accept any premium | 1-2 |
| H2 | Oji samples meet buyers' functional specs | Buyer-specified leak, grease and moisture tests | At least 90% pass rate | Below 70% | 2-4 |
| H3 | A viable landed cost exists | Cost model: import vs local production vs incumbent quotes | Landed cost within the premium from H1 | Above it on every route | 2-4 |
| H4 | Buyers will commit to a trial | Pitch pilot terms | At least 1 LOI or paid trial | No buyer commits | 5-6 |

The thresholds are my proposals, to be agreed with Oji's team.

**Budget (all assumptions, INR lakh):** samples and courier 2.0; functional tests 3.0; interview travel 1.5; regulatory and cost-model consultant 1.0; subtotal 7.5; plus 10% contingency gives 8.25 lakh (about INR 0.08 crore). This excludes Oji staff time and any capital expenditure.

## 8. Risks and unknowns

| Risk | Likelihood | Why it matters | Mitigation |
|---|---|---|---|
| Local incumbents undercut on price | High | Many Indian makers exist (S06, S07, S09-S12) | Landed-cost model against 2-3 incumbent quotes |
| No local supply route | High | S01 does not establish India availability; molded pulp is bulky | Test import vs local production or partner |
| Weak, conflicting market data | Certain | S07 and S08 cannot both be right | Replace with buyer-reported volumes |
| Claims unvalidated in Indian conditions | Medium | S01 is marketing only | Buyer-specified tests |
| Regulatory pull weaker | Medium | EPR is plastic-only (S04) | Ask buyers how EPR |

| Risk | Likelihood | Why it matters | Mitigation |
|---|---|---|---|
| than assumed | | | enters decisions |
| Food-contact compliance gap | Medium | Not researched here | Confirm with a regulatory consultant first |
| Unit economics fail despite demand | Low | Great Wrap failed with USD 39M of debt (S13), though it made compostable stretch wrap, not molded pulp | Kill criterion on unit economics |
| No primary data in this study | Certain | Desk research only | Interviews are step one |

## 9. Limitations

This report rests on 13 sources, of which two are low-reliability vendor summaries and five are company self-descriptions or trade press. I did not verify prices, capacities, certifications or customers. I did not find an official website for Ecology Pac, so that source is a marketplace listing. The NITI Aayog report was not read in full. The scores are judgements and are shown with their rationale so a reader can disagree and change them in the workbook.

## References

Access date for all web sources: 25 October 2026.


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